

Ultrasonic Prediction of Delamination in Cross-laminated Timber using CNNs

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Abstract

Cross-laminated timber (CLT) is a wood laminated composite consisting of planks of wood laid together in parallel within one lamella and glued to adjacent lamellas with their boards laid in the perpendicular orientation. This allows the construction of wooden elements larger than what can be harvested from solid timber, but it comes with its own considerations. One such consideration is that, due to the significant axial differences in wood dimensional change between the longitudinal and non-longitudinal directions, CLT bond lines can experience significant internal stress with added moisture content. CLT certification such as ANSI/APA PRG320 therefore specifies accelerated weathering analogous testing to ensure this bond quality. This study analyzes the ability of ultrasound scanning prior to delamination testing to predict its results without the need for the full test cycle.

1. Introduction

Cross-laminated timber (CLT) is a wood laminated composite consisting of planks of wood laid in parallel glued to adjacent lamellas with their boards laid in the perpendicular orientation. This is done to form wood-composites larger than that can be harvested from solid timber, but it comes with its own considerations. One such, is that due to the significant axial differences in wood dimensional change between the longitudinal and non-longitudinal directions (which alternate in orientation between lamellas in CLT), CLT bond lines can experience significant internal stress with variations in moisture content.

CLT standards, such as ANSI/APA PRG320 specify moisture cycling tests to simulate this long-term defect potential. These test are destructive and can only be conducted on a few samples to avoid manufacturing line disruption. As a more rapid and nondestructive alternative or precursor to the prescribed testing, this paper evaluates the ability of ultrasound through-transmission testing prior to the de-

lamination to predict the delamination using convolutional neural networks (CNNs).

Similar work has been done before on other layered wood composites including plywood and glulam [3, 2], looking at detection the deliberately compromised glue-line. More directly relevant, Sebara *et al.* [7] and Concu *et al.* [4] have found promising result in detecting deliberate bond quality issues in CLT. These studies primarily analyzed the signal in terms of time-of-flight and amplitude, which, while retaining a strong physical meaning, do ignore the depth of information that could be extracted from the reflections in the signal as a whole.

2. Methods

2.1. Sample Preparation

Standardized testing to simulate the affects of long term-moisture cycling consists of subjecting small block specimens cut from the larger panel to a delamination cycle. For this dataset, the testing was conducted according to ANSI/APA's PRG 320-2025[1]. This includes vacuum and pressure cycles while being soaked in water and rapid kiln drying immediately before delamination assessment. The assessment consists of measuring lengths of adhesive attributed delaminations along the edge of the bond-lines and presents it findings as the total linear delamination over the total bond edge length on the block, as shown in equation (1).

$$AF = \frac{\text{sum of edge delamination lengths}}{\text{sum of edge lengths}} \times 100 \% \quad (1)$$

Delamination is also sometimes evaluated as the area of the bond-line where failure can be attributed to the adhesive after cracking the sample open along the bond-line with a chisel. Both metrics are investigated in this report, the first being referred to as *edge*, and the second as *area*.

This delamination percentage is the classified into pass and fail based on the magnitude of the delamination (al-

though PRG 320 does allow for a marginal case for specimens that are close to passing.) For this study, passing is considered as below 5 % adhesive failure.

2.2. Dataset

The ultrasonic testing was conducted using Pundit Lab's PL-200 ultrasonic pulse velocity tester, which emits and captures ultrasound signals with paired transducers. For this study, the accompanying 54 kHz transducers were used at an excitation of 400 V with gains adjusted between $\times 5$ and $\times 200$ to amplify the received signal into the ± 2 V range of the ADC. The transducers were applied in conjunction with a spring controlled enclosure to ensure a consistent application pressure of 17 kPa, and no couplant was used.

Tests were performed on twelve lab-made CLT layups in both 3 ply and 5 ply. Each layup was cut into three specimens for a total of 72 blocks. Each of these blocks were tested with the ultrasound device 50 times to get 3600 total signals each 7928 samples long. However, as there are only 72 unique blocks and only 72 delamination values, this is a small sample size for deep learning, such as CNNs.

In an effort to partially combat the imbalance between ultrasound signal data and the resulting delamination data, delamination results were duplicated with realistic measurement error noise. This was applied at the delamination length level, before being derived into a percentage using equation (1) on the previous page with an estimated normal with of 1 mm per measurement (on the approximately 80 mm long edges).

Furthermore, to combat the overall small sample size, the 3 ply and 5 ply data were combined, while providing models with a data marker for the number of layers present in the sample. This was done under the assumption that defect correlated signal features might be small scale occurrences within the overall signal, and therefore be less affected by the difference in time-of-flight present in the signal. This was confirmed in preliminary testing where combining the data slightly improved the model.

In addition, a second small dataset was provided as more training data. This dataset was produced under a less controlled environment, but with the same equipment and similar specimens.

All evaluation is performed in six-fold k-fold cross validation, with the fold splits carefully selected such that these conditions are met:

1. Blocks from the same layup are always within the same fold.
2. An equal number of 3 ply and 5 ply layups are present in each fold.
3. And each fold contains at least one pass and one failure example for both 3 ply and 5 ply.

An arbitrary fold order was found that accommodated these restrictions and was used for all results reported.

2.3. Modeling

Two different general model architectures were evaluated for this dataset, as can be seen as the left and center diagrams in figure 1 on the following page.

1. A simple chain of convolutions with interconnected residuals on the 1D signal (1DCNN) and with the fast Fourier transform (FFT) of the signal (1DCNN+).
2. Resnet-18 encoder with small depths on the spectrogram of the signal. The Resnet-18 encoder is pre-trained for spectrogram reconstruction on unclassified ultrasound signals from this device, before being transferred to this model (2DCNN).

All models were terminated by several linear layers with a dual value output for binary focal cross-entropy loss. This is in an attempt to mitigate the effects of a class imbalance, where most blocks pass the delamination test. Models will be evaluated in terms of their F1 score and AUROC score.

These approaches will be compared against a non-deep learning approach using a scalar vector machine with an RBF kernel similar on metadata from variable mode decompositions of the signal (SVC), similar to an approach by Mousavi *et al.* [6] on the same equipment with similar specimens. They will also be compared against a trivial linear neural network, where all of the features are combined without any convolution (Linear).

3. Results

No model is able to achieve significant improvements over random guessing in the test data (as shown in figure 2 on the next page), although models fitting for *area* delamination are consistently more performant than models fitting for *edge* delamination. Physically, this could make sense, as the bulk of the wood the ultrasound signal interacted with was centered in the middle of the area, while the edges likely would have little to no direct affect on the signal. With the small sample size, the small difference present is not enough to make a definitive observation on the matter, however.

Also of note is the behavior of Linear compared to the CNNs and SVM. The edge prediction was randomly guessing like the rest of the models, but the area prediction managed the highest F1 score at 0.62 while keeping its AUROC score at around 0.5. This likely indicates a breakdown in the model, and a failure to recognize anything beyond predicting the majority class all of the time.

This can be seen by looking at the class-wise accuracy in figure 3 on the following page. Linear pass every time, as

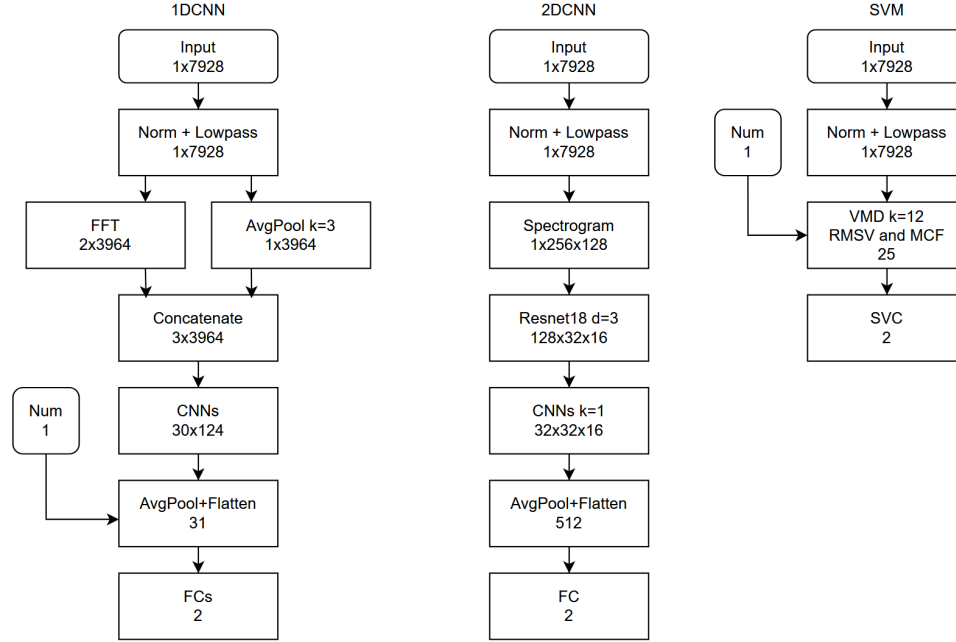


Figure 1. Three diagrams detailing the general data flow for each model used in this study. The left diagram details the 1D convolutional neural network on the signal and its fast Fourier transform (FFT) (1DCNN+), the center diagram details the 2D CNN on the signal spectrogram (2DCNN), and the right diagram details the flow used for the comparison support vector machine (SVC).

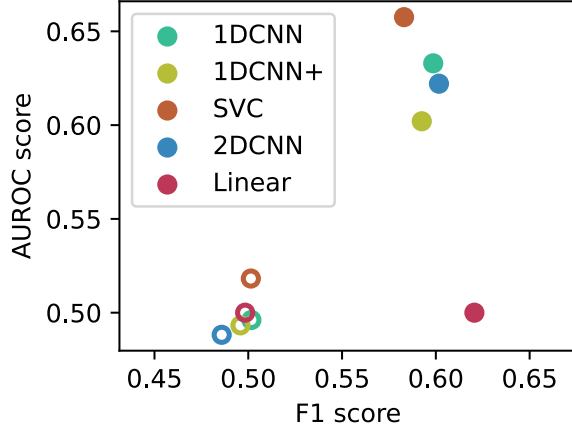


Figure 2. A scatter plot comparing F1 score and AUROC score between the models evaluated. Hollow markers indicate models trained predicting edge delamination, while filled markers indicate models trained predicting area delamination.

that is the majority class, with F1 just scoring its confidence thresholds on the relevant classes.

4. Conclusion

No deep learning example was able to achieve any significant improvement in accuracy over SVC. At best they remained capped at a F1 score of around 0.6, indicating lit-

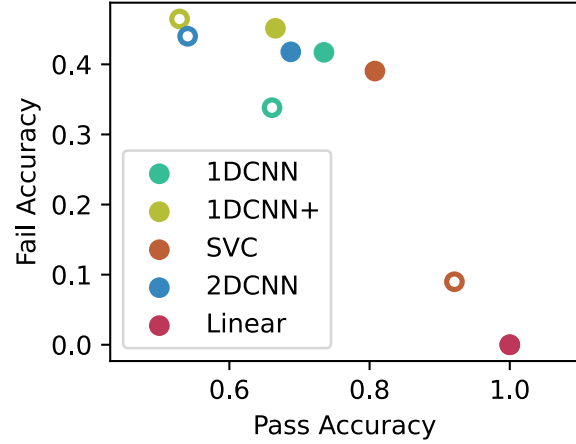


Figure 3. A scatter plot comparing class-wise accuracy between models. The majority class, pass, is on the horizontal axis, while the minority class, fail, is on the vertical. Note the bottom right Linear point is both edge and area predictions overlapping. Hollow markers indicate models trained predicting edge delamination, while filled markers indicate models trained predicting area delamination.

tle to no learning. Models complicated enough to internalize the signal data, also seem to readily over-fit to it. The models tested only achieved a marginal accuracy, despite restraining tactics such as class balancing, and focal loss.

As well as attempting to augment the data in the model by pre-training the encoder for 2DCNN and adding in data from practice samples to the training.

This report would attribute those failures to the lack of sufficient varied data. While this report has gone through a great variety of techniques to address this that were at best marginally successful, there remain a number of potential solutions for future investigation:

- Gather more data. While it is difficult to produce CLT elements, manufacturing facilities are required to conduct periodic tests of their samples from the production line. As the ultrasound measurements are nondestructive, integrating in measurements before planned testing could be an easy (if slow) way to gather more data from a greater variety of specimens.
- Another direction could be with physics-guided learning. With a sophisticated enough model, calibrated to produce simulated ultrasonic measurements for a known variety of defect qualities, artificial samples could be added into the training data or used for pre-training to direct the complex models to the relevant portions of the ultrasound signals. This has been shown to be fairly successful in similar ultrasonic defect detection [5].

The more consistent performance of the non-linear model indicates there is some level of shape to the signal features that is being learned, but it does not lead to a significant prediction ability in this study. Nor does it seem to favor CNNs over signal decomposition in conjunction with support vector machines in this dataset.

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